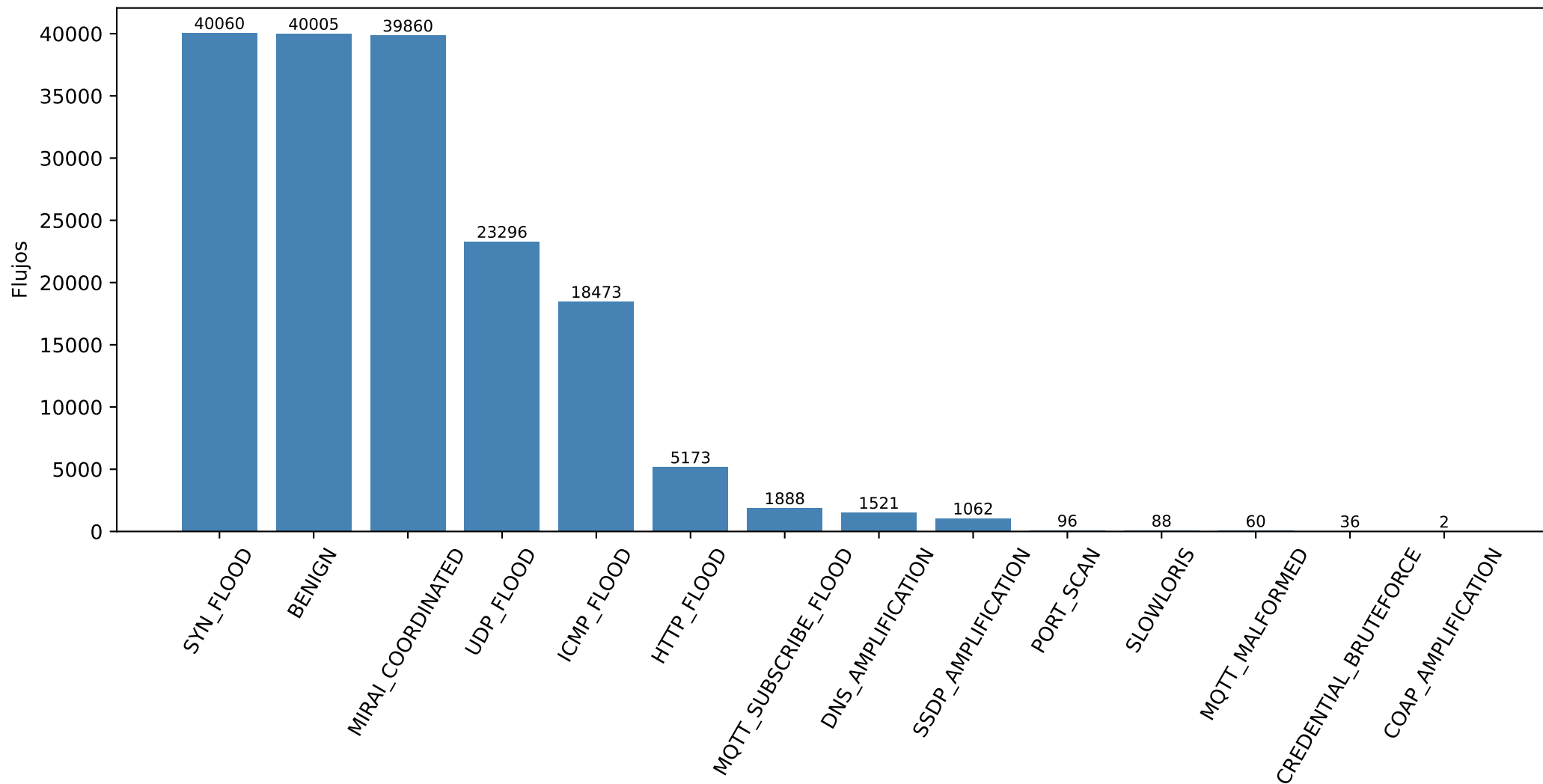


IoT-SDN-AI — Reporte de evaluación

run: run_20260921-135634

Dataset:	208,137 flujos
Features:	46
Clases:	14 (BENIGN, COAP_AMPLIFICATION, CREDENTIAL_BRUTEFORCE, DNS_AMPLIFICATION, HTTP_FLOOD...)
Modelos:	RandomForest, XGBoost, MLP, IsolationForest, AutoEncoder
Estrategia:	grouped split anti-fuga (StratifiedGroupKFold fold0) 80/20, StandardScaler

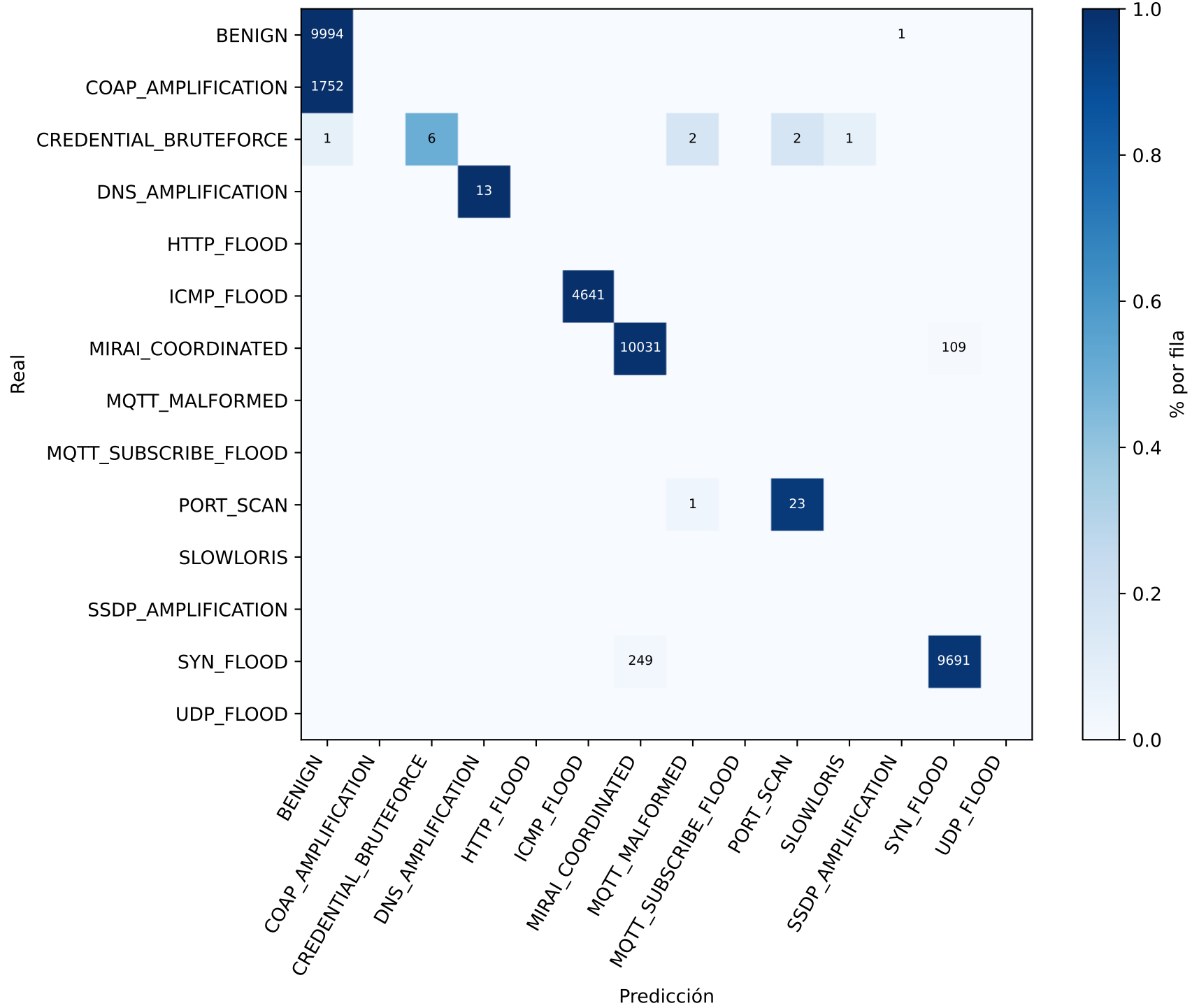
Distribución de clases en el dataset



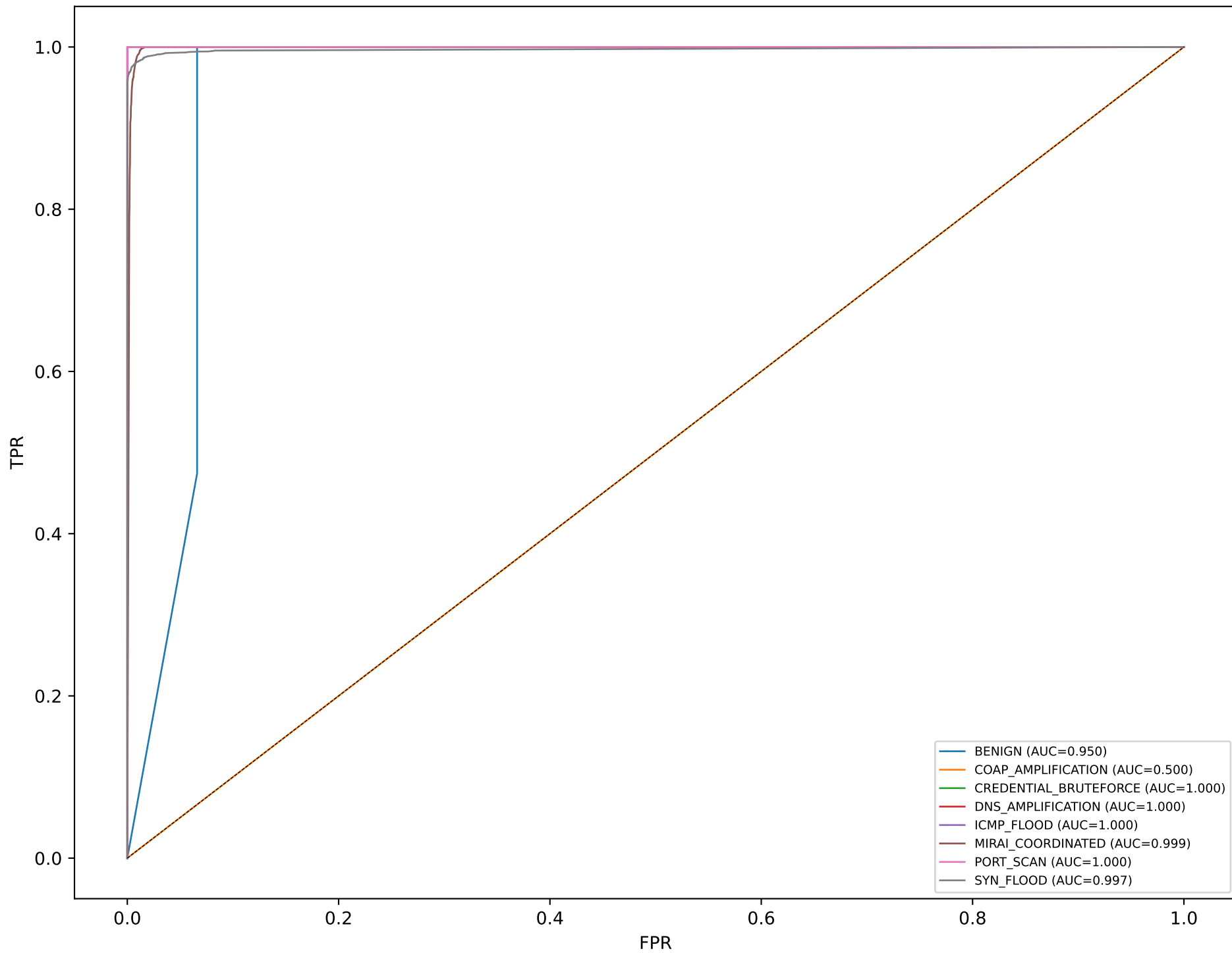
Classification report — rf

	precision	recall	f1-score	support
BENIGN	0.85	1.00	0.92	9995
COAP_AMPLIFICATION	0.00	0.00	0.00	1752
CREDENTIAL_BRUTEFORCE	1.00	0.50	0.67	12
DNS_AMPLIFICATION	1.00	1.00	1.00	13
ICMP_FLOOD	1.00	1.00	1.00	4641
MIRAI_COORDINATED	0.98	0.99	0.98	10140
MQTT_MALFORMED	0.00	0.00	0.00	0
PORT_SCAN	0.92	0.96	0.94	24
SLOWLORIS	0.00	0.00	0.00	0
SSDP_AMPLIFICATION	0.00	0.00	0.00	0
SYN_FLOOD	0.99	0.97	0.98	9940
accuracy			0.94	36517
macro avg	0.61	0.58	0.59	36517
weighted avg	0.90	0.94	0.92	36517

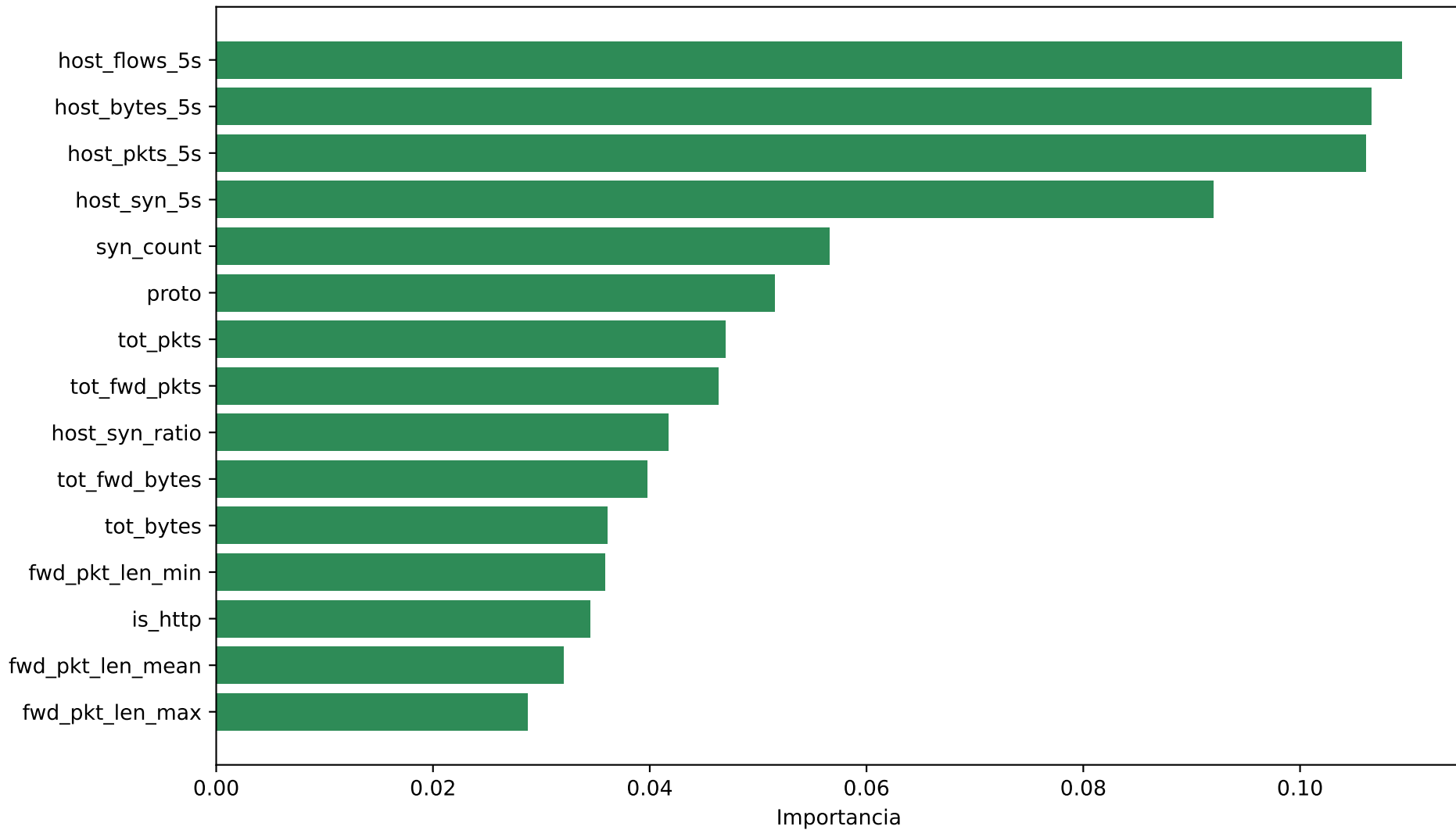
Matriz de confusión — rf



ROC OvR — rf



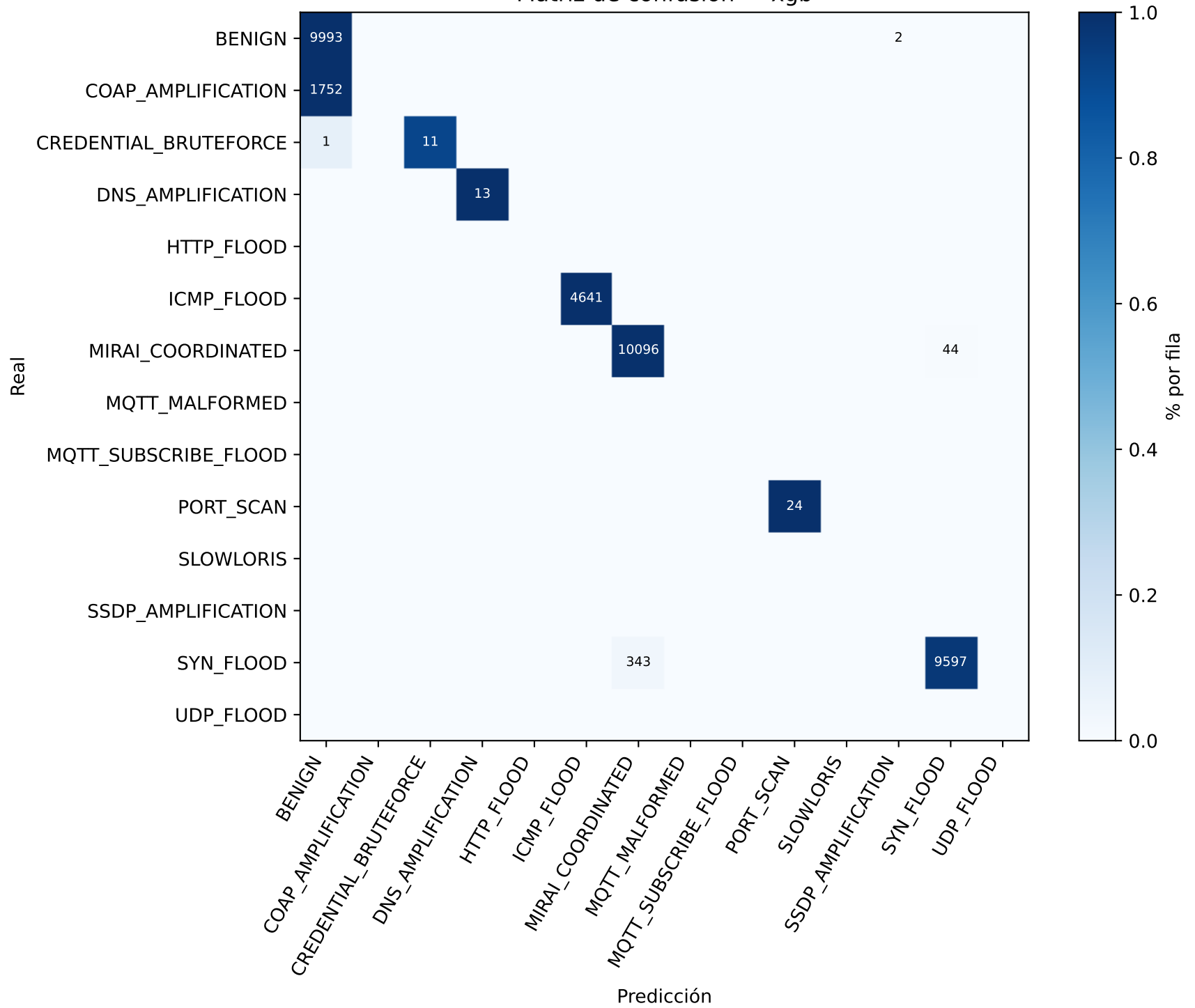
Top-15 features — rf



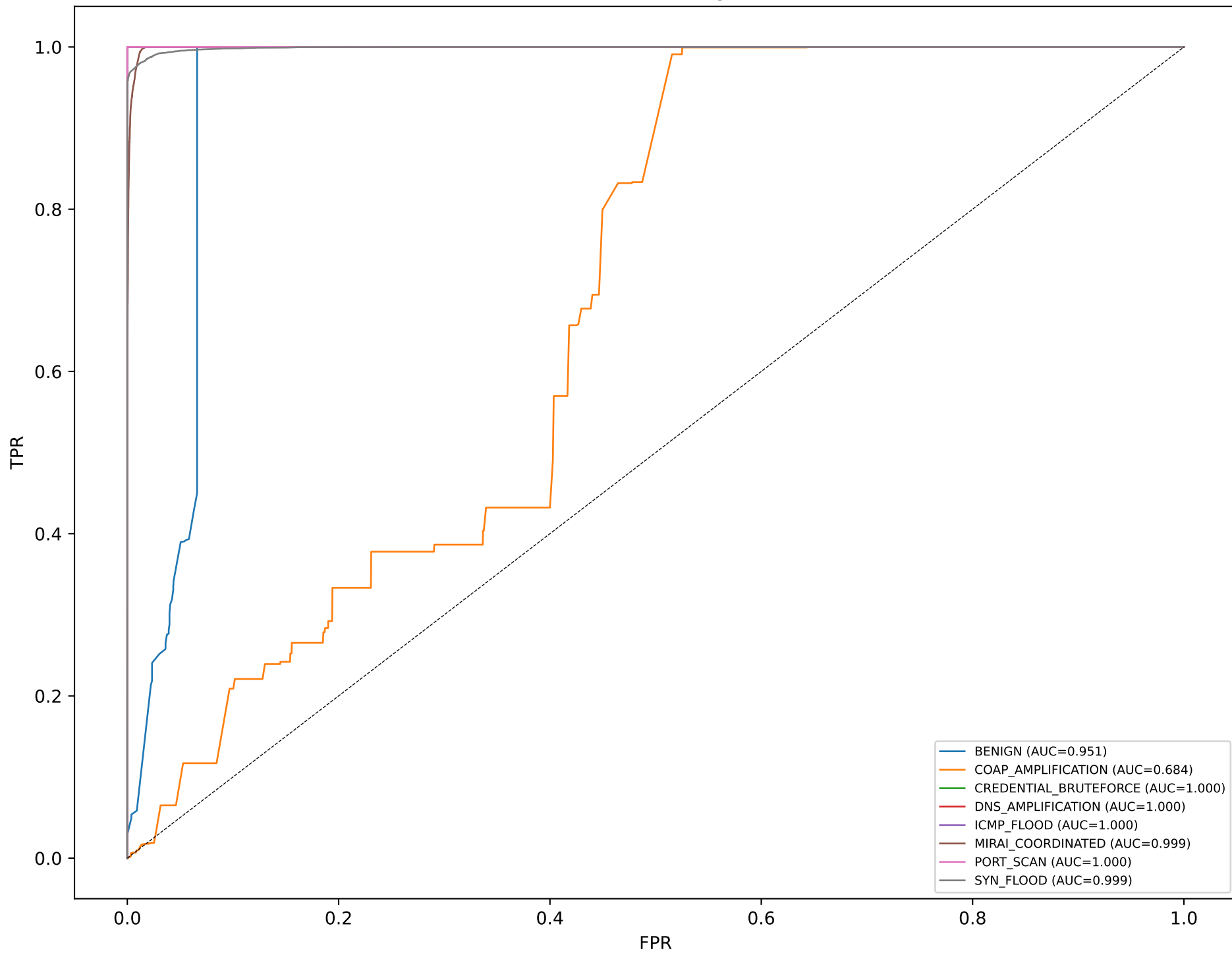
Classification report — xgb

	precision	recall	f1-score	support
BENIGN	0.85	1.00	0.92	9995
COAP_AMPLIFICATION	0.00	0.00	0.00	1752
CREDENTIAL_BRUTEFORCE	1.00	0.92	0.96	12
DNS_AMPLIFICATION	1.00	1.00	1.00	13
ICMP_FLOOD	1.00	1.00	1.00	4641
MIRAI_COORDINATED	0.97	1.00	0.98	10140
PORT_SCAN	1.00	1.00	1.00	24
SSDP_AMPLIFICATION	0.00	0.00	0.00	0
SYN_FLOOD	1.00	0.97	0.98	9940
accuracy			0.94	36517
macro avg	0.76	0.76	0.76	36517
weighted avg	0.90	0.94	0.92	36517

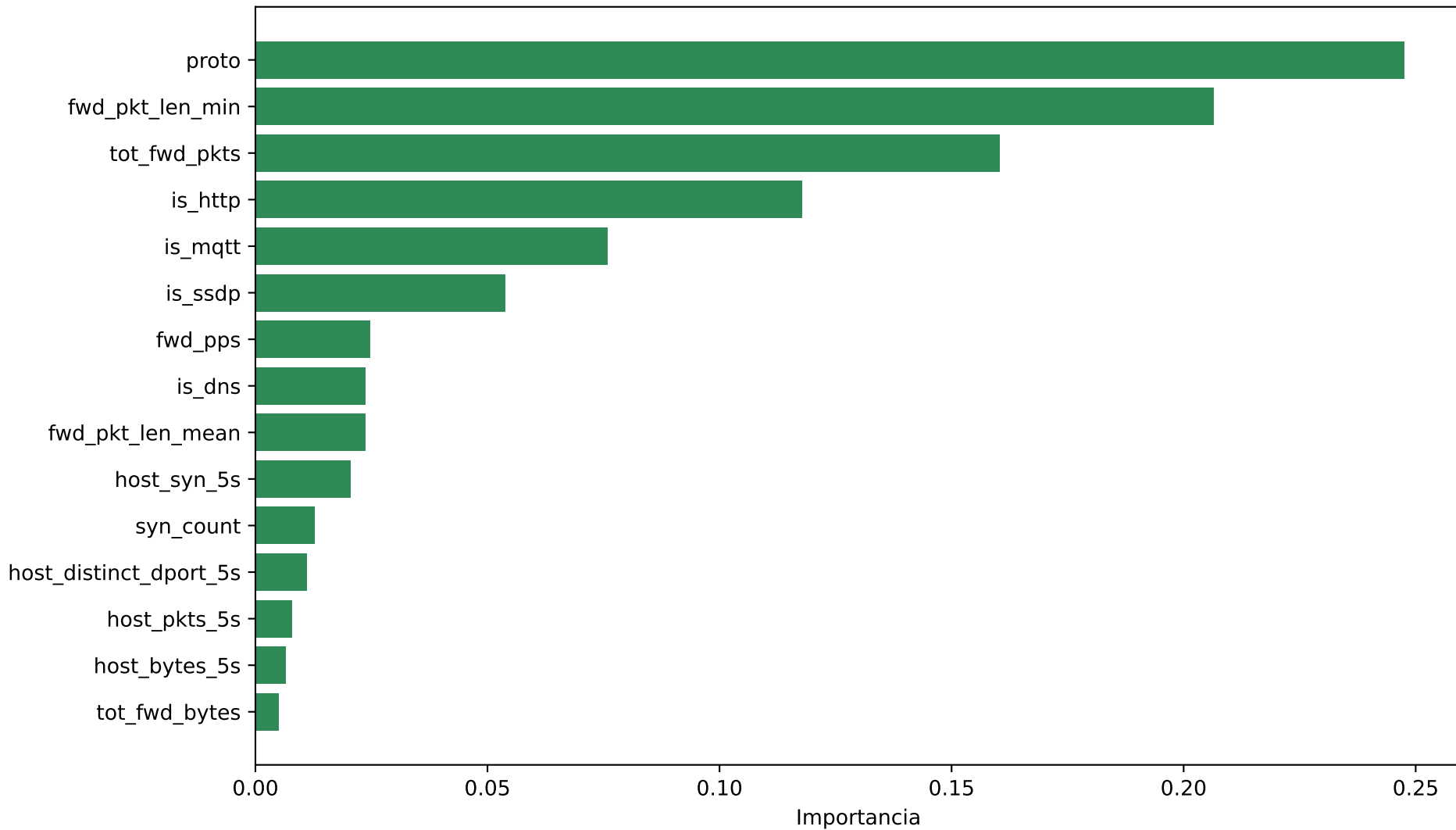
Matriz de confusión — xgb



ROC OvR — xgb



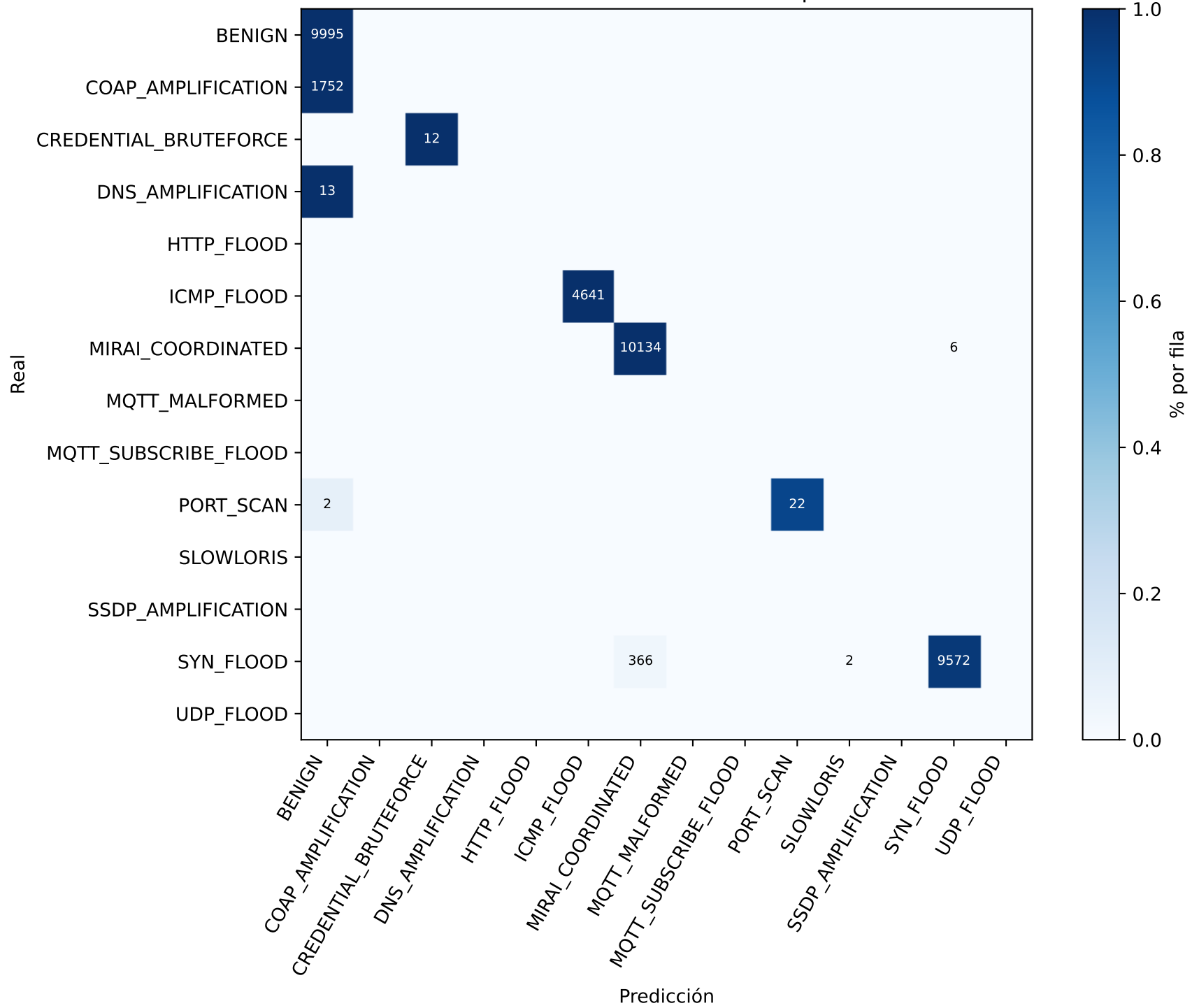
Top-15 features — xgb



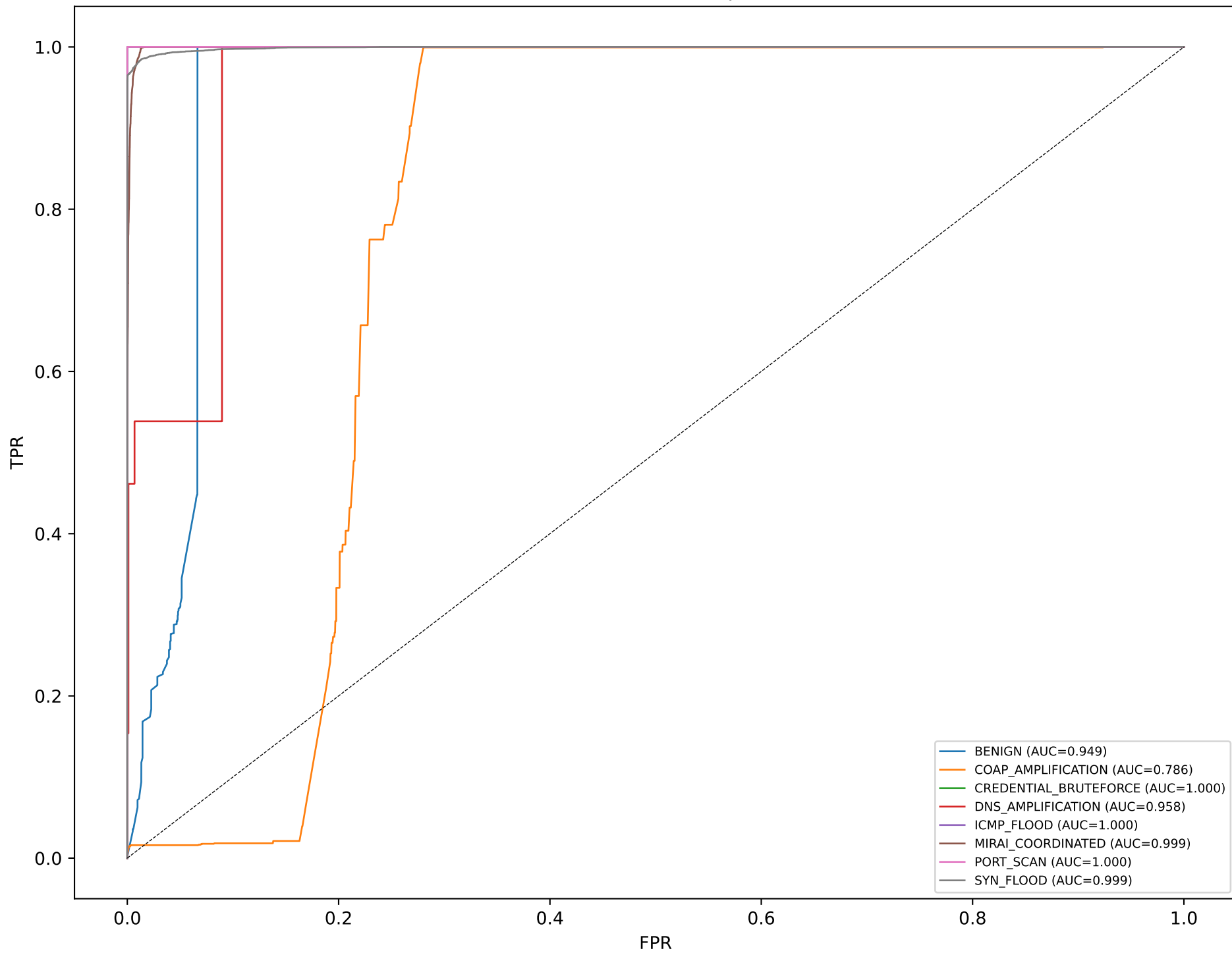
Classification report — mlp

	precision	recall	f1-score	support
BENIGN	0.85	1.00	0.92	9995
COAP_AMPLIFICATION	0.00	0.00	0.00	1752
CREDENTIAL_BRUTEFORCE	1.00	1.00	1.00	12
DNS_AMPLIFICATION	0.00	0.00	0.00	13
ICMP_FLOOD	1.00	1.00	1.00	4641
MIRAI_COORDINATED	0.97	1.00	0.98	10140
PORT_SCAN	1.00	0.92	0.96	24
SLOWLORIS	0.00	0.00	0.00	0
SYN_FLOOD	1.00	0.96	0.98	9940
accuracy			0.94	36517
macro avg	0.65	0.65	0.65	36517
weighted avg	0.90	0.94	0.92	36517

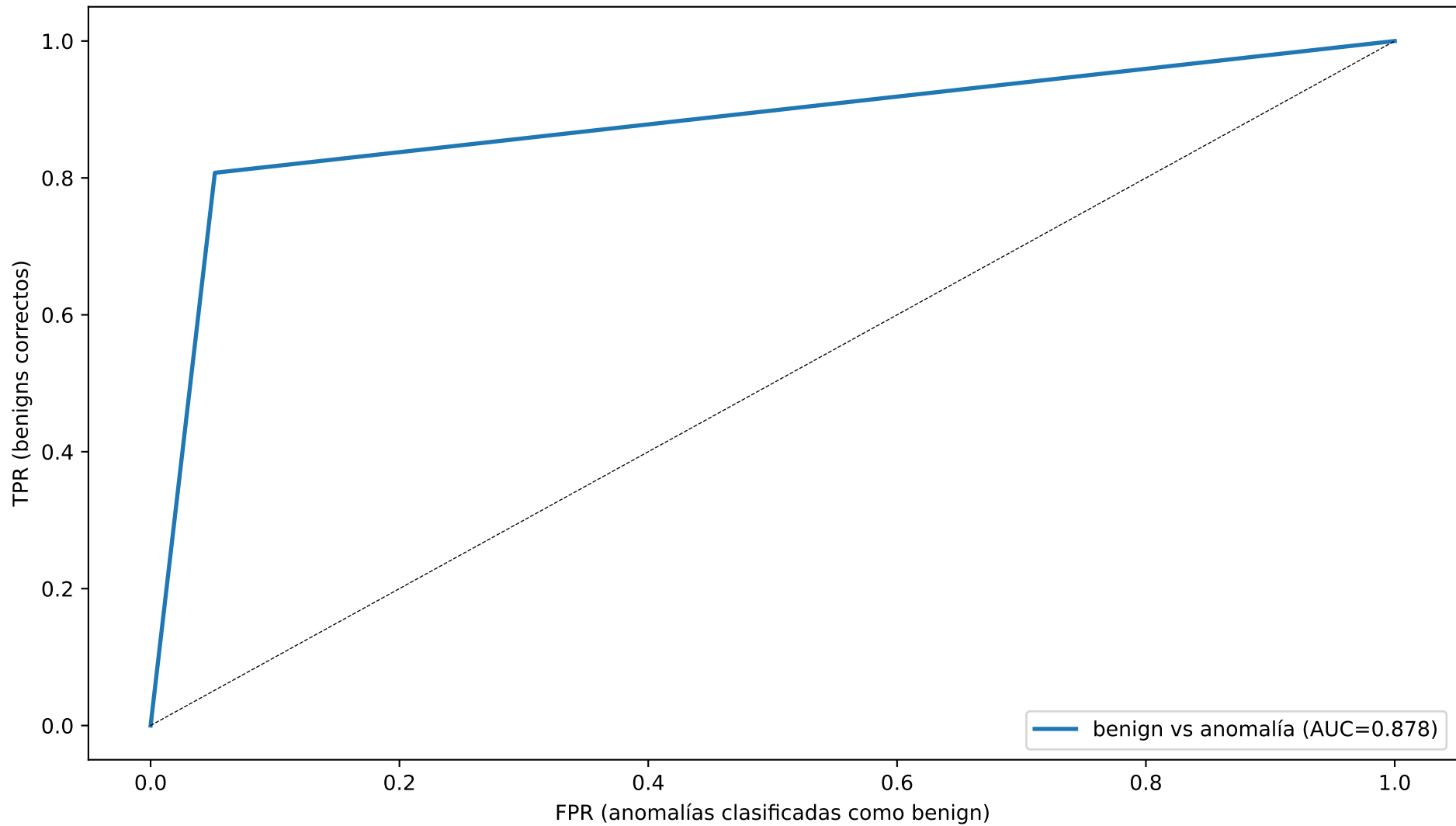
Matriz de confusión — mlp



ROC OvR — mlp



ROC binaria — isoforest



ROC binaria — autoencoder

